SMART OPERATIONS / DIGITAL MINE

SmartHaul

Digital Mine Scenario Walkthrough

Interactive prototype for open-pit haulage scenarios

Six scenarios

Overview - Truck cycle - Queue - Road closure

Vehicle breakdown - Scenario comparison

Contest prototype | Synthetic data | Not connected to live equipment

01 / CONCEPT

A clearer view of haulage operations

SmartHaul is a demonstration concept for a digital open-pit mine.

It helps explain how local disruptions affect routes and truck flow.

Operational scenarios

01 Mine overview and road network

02 Truck cycle: loading, haul, unload, return

03 Queue at loading point

04 C-3 road closure and illustrative detour

05 HT-04 breakdown; other trucks continue

06 Compare illustrative network states

The page is a visual explainer, not a dispatch control system.

02 / IMPLEMENTATION STATUS

Prototype boundaries

Included

Standalone Canvas scene and six navigable scenarios

Pause, reset, keyboard navigation, drag and zoom

Documentation of assumptions and known limitations

Not yet validated

Live MQTT / Eclipse Ditto integration

Production dispatch optimization or real savings

Clean-environment integration tests

All metrics in the demo are synthetic and illustrative.

Do not use this prototype to control equipment or for safety decisions.